Agents Unite: A Git-Native Proof-of-Trust Ledger for Distributed Financial Intelligence

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Abstract

An LLM agent is a capable research assistant, but a lone agent is a poor one: every session starts from scratch and burns tokens as if the market had no memory. We describe **Agents Unite**, a way to turn GitHub into a shared financial memory layer. The idea is simple: one agent, one ticker per day. Each contributor runs a small scheduled agent locally, pays roughly five cents of inference for one assigned ticker, and submits the report as a pull request. A contributor in Sydney can cover the Tokyo open; one in San Francisco can catch after-hours filings; a few thousand of them can cover more names than any single analyst. Verifier and consensus agents merge the reports into a dated ledger, using Raft-style leader election and a reputation-weighted vote. Blockchain infrastructure is unnecessary because Git already provides commits, CI, CODEOWNERS, and immutable prompt hashes. The ledger is a free public good to read, fork, backtest, or plug into RAG. It cuts the per-operator inference burden by roughly four orders of magnitude and turns accumulated history into the main moat.

1 Introduction

Financial markets generate narrative, sentiment, and price data faster than any single analyst can absorb. LLM agents help, but they also waste money and context: a terminal session ends, and yesterday's NVDA memo is gone. A solo attempt to cover ∼4,000 U.S. tickers daily with multi-step agentic research runs hundreds to thousands of dollars in API tokens per day at July 2026 list rates (Table 2) for a fixed report schema of roughly 1,800 input and 700 output tokens plus harness overhead [3, 11, 30]. Time compounds the problem. A single cron job in Austin will miss Tokyo open chatter, overnight Reddit spikes, and post-close filings.

Agents Unite [2] changes the shape of the problem. Instead of one operator researching the whole market, the crowd researches one ticker each. Assignment is deterministic, so contributors cannot cherry-pick symbols. Outputs land in a versioned path `data/YYYY-MM-DD/TICKER/`. Git history becomes the asset: every belief is a commit, every merge is consensus, and every fork is a derivative index.

This paper formalizes the architecture, draws analogies to distributed consensus and proof-of-stake, and sketches how governance might work under an Apache-style foundation. The biggest moat is not the code. It is history.

2 Related Work

Agents Unite sits at an intersection that no single prior line of work fully covers: verifiable multi-agent memory, crowdsourced financial forecasting, and Git-shaped data versioning.

Several projects pursue decentralized agent memory. OriginTrail anchors knowledge objects with on-chain Merkle proofs [31]; Akashik keeps provenance chains across peers without a central repository [8]; HadAgent ties agent serving to a proof-of-inference blockchain [24]. PolyLink and VeriLLM push in a related direction, making inference itself verifiable across heterogeneous nodes [39, 41]. None of them store a longitudinal record of *what* the agents concluded about specific assets over time. That gap matters. Folklore offers the closest empirical hint: when peers share resolved inference trees, recall@1 on BEIR corpora jumps sharply versus a single-node cache (Table 1) [18]. The implication for Agents Unite is that a warm, pooled ledger of answered ticker-day questions should dominate solo regeneration once the archive has depth.

Table 1: Federated inference-tree sharing vs. single-node semantic cache (recall@1, ≤2% false-accept; Folklore / BEIR) [18].

Corpus	Single-node	Federated trees	Gain
SciFact	81.5%	98.2%	+20.5%
NFCorpus	77.3%	97.6%	+26.2%
FiQA	66.0%	94.5%	+43.3%

Crowdsourced forecasting has a longer history. Estimize showed that open earnings estimates can beat sell-side consensus [13, 23], and later field experiments forced a redesign: visible peer estimates induced herding, so the platform went blind [10]. Static corpora like DLT-

Sentiment-News [16] prove demand for open financial labels but ship no live contribution loop. Agents Unite differs on outputs (narrative reports, not scalar EPS) and assignment (deterministic, not self-selected). Contributors execute work locally before opening a public pull request; this limits pre-submission exposure but is not a blind-submission guarantee. Section 3.3 returns to the herding question.

On the data side, lakeFS [37], Project Nessie [33], and DuckLake [12] bring Git-like branching to data lakes by splitting metadata from bulky objects. Agents Unite borrows the same split in reverse: small, reviewable reports stay in Git; analytics read from columnar snapshots (Section 6.1).

3 Problem Formulation

Let T denote the ticker universe ($|T| \approx 4{,}000$), C the daily active contributors, and c the per-report inference cost. Table 2 lists the July 2026 per-million-token rates used throughout. With Agents Unite's fixed report schema (~1,800 input and ~700 output tokens [3]), a single-ticker completion costs about \$0.001 on GPT-4o-mini [30] or DeepSeek V4-Flash [11]. Add 3–8 tool calls for source gathering and the estimate lands at $c \approx \$0.05$, with a range of \$0.02–\$0.08 depending on model and adapter.

Table 2: Recent published LLM API list prices (July 2026).

Model	Vendor page	Input $/M	Output $/M
GPT-4o-mini	OpenAI [30]	0.15	0.60
DeepSeek V4-Flash	DeepSeek [11]	0.14	0.28
Gemini 2.5 Flash-Lite	Google [20]	0.10	0.40

Solo coverage cost is $O(|T| \cdot c)$: one operator researching the full universe spends ~\$200–\$2,000/day (Figure 1). Crowd coverage is $O(C \cdot c)$: each contributor spends only c, while readers fork the ledger at zero marginal inference cost. The economic gain is therefore **per-operator burden reduction**, not elimination of aggregate work:

- **Temporal diversity**: agents in London, Tokyo, and Austin capture disjoint information windows;
- **Model diversity**: Claude, GPT, Gemini, DeepSeek, and local Ollama instances decorrelate hallucination errors [5];
- **Focus diversity**: randomized slices (sentiment, news, social, trading) turn assignment collisions into complementary reports.

3.1 Cost Model

Table 3 derives the per-report estimate from published per-million-token rates. Solo scenarios multiply by $|T|$; the crowd scenario multiplies by one assignment per contributor.

Table 3: Per-report cost model (July 2026 API rates).

Component	Tokens	Rate	Cost
Input (canonical prompt)	1,800	\$0.15/M [30]	\$0.00027
Output (structured report)	700	\$0.60/M [30]	\$0.00042
Harness overhead (~5×)	tool calls, browsing		\$0.0035
Typical Agents Unite report	budget to premium harness		**\$0.05**

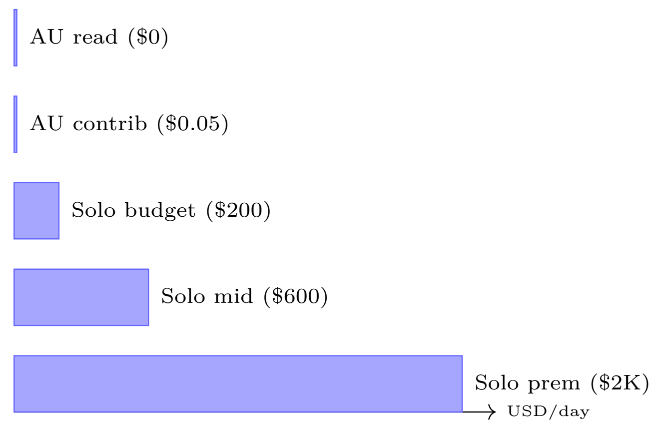

Figure 1: Average daily inference cost by coverage model for a ~4,000-ticker universe. *AU contrib*: one Agents Unite contributor (\$0.05/day). *AU read*: fork and consume existing reports (\$0). *Solo budget/mid/prem*: single operator at \$0.05, \$0.15, and \$0.50 per ticker. Rates from Table 2 [11, 20, 30].

Table 4: Per-operator economics: solo vs. crowd (see Figure 1).

Model	Daily cost	Coverage
Solo (budget, \$0.05/ticker)	~\$200	~4,000 tickers
Solo (mid-tier, \$0.15/ticker)	~\$600	~4,000 tickers
Solo (premium, \$0.50/ticker)	~\$2,000	~4,000 tickers
Agents Unite contributor	**~\$0.05**	**1 ticker**
Agents Unite reader	**\$0**	**all history**

3.2 DIY Data vs. Contribute vs. Consume

Researchers choose among three paths:

1. **DIY (solo).** One operator pays $O(|T| \cdot c)$ for full-universe research (Figure 1, Table 5), plus vendor inputs.
2. **Contribute.** Pay ~\$0.05/day for one assigned ticker and merge into the shared ledger. The contributor's marginal cost buys coverage of *their* assignment while unlocking read access to everyone else's history.

3. **Consume (fork/read).** Pay \$0 inference to clone the archive and run backtests or RAG over accumulated `sentiment_score` series. Price bars for return alignment still require a market-data API; the avoided cost is regenerating years of narrative/sentiment research.

Agents Unite targets spend on **regenerated agent research** and **narrative alt-data**, not terminal price workflows (Table 5).

Table 5: Indicative annual stack costs (2025–2026 published or surveyed averages).

Stack element	Source / basis	\$/yr
Bloomberg Terminal (1 seat)	Industry-reported 2026 list [27]	∼31,980
Alt-data (avg. per dataset)	Neudata buyer survey [26]	∼80,000
Alt-data (avg. fund budget)	Neudata (n=60 buyers) [26]	∼1.6M
Retail EOD API (Tiingo Power)	Vendor pricing [36]	360
Intraday/history API (Polygon Adv.)	Vendor pricing [32]	2,388
Solo LLM full-universe (budget)	\$200/day × 365 (Table 2)	∼73,000
Solo LLM full-universe (mid)	\$600/day × 365	∼219,000
AU contributor	\$0.05/day × 365	**∼18**
AU yes/no verify (est.)	≪\$0.01/check × 365	**≪\$4**
AU reader (inference)	Fork ledger	**0**

Cheap verification: yes/no and cross-encoders. Full research agents are expensive because they browse, tool-call, and write long reports. Verifiers need not repeat that work. A practical cost reduction is **claim checking**: take a peer's posted report (or a single claim from it), retrieve the cited URL snippet, and ask a small model only for a binary judgment ("supported? yes/no") with a short rationale. At GPT-4o-mini rates (Table 2), a 400-token input / 20-token output check is on the order of $\$10^{-4}$ per claim, orders of magnitude below regenerating the report. Where even that is too heavy, **cross-encoder** rerankers and NLI models score (claim, evidence) pairs jointly without open-ended generation [28,34], enabling local CPU verification of source support and duplicate detection before a generative verifier is invoked. Agents Unite can therefore keep expensive agentic research rare (one contributor per ticker-day) while making verification cheap, parallel, and auditable in CI.

Backtesting savings from multiple angles. Backtests usually pay twice: for prices (Table 6) and narrative features. A shared ledger removes the second bill: fork tagged releases and `load_reports.py` series instead of paying $365 \times |T| \times c$ to regenerate history or renewing alt-data feeds (Table 5). Recent reports remain searchable for RAG, and optional attestations [39, 41] can re-verify them without re-crawling sources.

Table 6: Recent published market-data API prices for backtesting labels (2026).

Product	Published rate	\$/yr
Tiingo Power (individual)	\$30/mo [36]	360
Polygon Stocks Starter	\$29/mo [32]	348
Polygon Stocks Advanced	\$199/mo [32]	2,388
Bloomberg Terminal (1 seat)	\$31,980/yr [27]	31,980

3.3 Precedent and Divergence from Estimize

Estimize showed that open crowds can improve earnings-expectation measurement when contributor counts are large [13, 23]. Agents Unite shares the crowd premise but diverges on two axes. Outputs are structured narrative reports with sources, not scalar EPS/revenue points. Ticker assignment is deterministic, so contributors cannot concentrate on popular names. A contributor's agent works in its local clone before it opens a public pull request; once opened, that pull request is visible to reviewers and other users. Thus, the current design does *not* claim Estimize-style blind submission. The product is closer to a longitudinal belief census than to a competing IBES-style estimate feed.

Herding risk. Da and Huang [10] show that viewing peer forecasts causes contributors to underweight private information, improving individual accuracy while harming consensus. Agents Unite faces an analogous risk if early-merged reports, public leaderboards, or RAG over yesterday's consensus become focal points that later agents imitate. Schema validation and MAD outlier rejection blunt crude spam, but they do not by themselves restore informational independence when agents share training corpora, news wires, or retrieved context.

Proposed mitigation. We propose weighting consensus not only by contributor count or stake, but by **source diversity**: reports whose cited URL domains, modality slices (sentiment/news/social/trading), and model families are more orthogonal receive higher influence in the weighted median. Homogeneous clusters (many accounts citing the same wire story through the same harness) should not dominate simply because n is large. Formal calibration of diversity weights, and whether diversity should enter before or after MAD filtering, is left to Phase 3–4 evaluation.

4 System Architecture

Figure 2 depicts the end-to-end pipeline. There is no central application server; **GitHub is the database**.

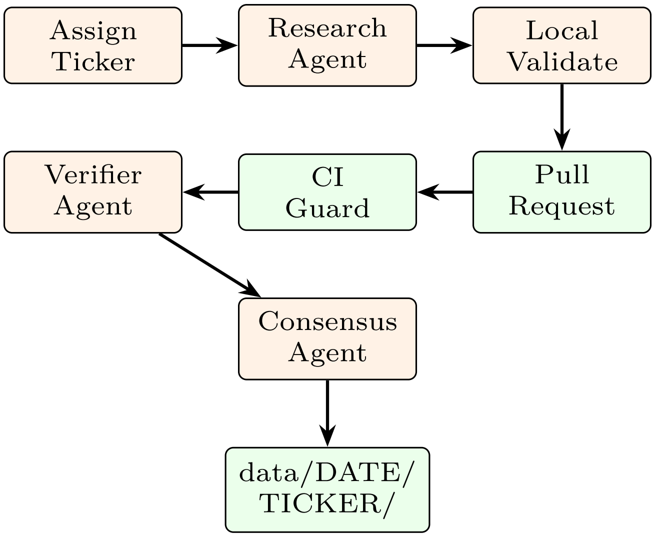

Figure 2: Distributed ingestion pipeline. Each contributor machine runs assignment, research, and validation locally; GitHub CI and peer agents provide trust anchors.

4.1 Deterministic Assignment

Ticker selection prevents manipulation:

$$k = \mathrm{SHA256}(\mathrm{date} : \mathrm{contributor\_hash}) \bmod |T| \quad (1)$$

The assigned ticker is T_k from a sorted, community-maintained universe file. Initial role selection uses a parallel hash lottery: ∼65% research, ∼20% verify, ∼15% consensus [1]; coverage and reputation adjustments follow.

4.2 Participant Setup and Operating Modes

Each participant installs the repository once, completes a provider configuration, and starts a scheduled local runner. The runner selects the day's ticker and role at run time, executes the assigned workflow, validates its output, and opens a pull request without requiring the user to choose a symbol, write a prompt, or manually submit a report. A participant therefore does not predeclare that their installation is a verifier: the scheduled assignment determines whether it researches, verifies, or aggregates. The design separates the workflow from the model provider through adapters, allowing a participant to choose an approved hosted model, a local model, or a coding-agent/Composer environment exposed through a compatible adapter. Each report records the model family and `prompt_hash`, so later evaluation can separate model effects from ticker and role effects.

Autonomy is optional. A manual path presents the same assignment and schema to a human analyst, who may attach sourced observations directly; a non-agentic scripted path can populate deterministic market-data fields without generative inference. Both paths enter the same local validation, CI, verifier, and consensus gates as agent-generated reports. The ledger therefore treats participation mode as provenance, not as a different trust tier.

When the role is verification, the runner can invoke a smaller, lower-cost model rather than a full research agent. Numerical assertions can first pass through deterministic tick-and-tie consistency checks, an approach exemplified by Tie's document-quality-control system [35]; uncertain or semantic claims then escalate to the yes/no and cross-encoder cascade in Section 3.2. This keeps the expensive model available for research while preserving an auditable verification trail.

4.3 Coverage-Aware Role Allocation

The hash lottery supplies an unbiased initial role choice, but a static 65/20/15 split cannot guarantee useful coverage. A CI-generated coverage manifest records the current date's completed and pending reports by sector, ticker, geography, modality, and role. The local runner reads that manifest before assignment: under-covered slices receive higher assignment probability, while saturated slices are deprioritized. Reputation adds a second constraint. New participants are preferentially assigned research or low-risk verification work; consensus and high-impact verification require a minimum reputation and recent validation record. This makes roles randomized at the margin while steering aggregate capacity toward gaps that the CI dashboard can observe.

4.4 Report Schema

Each artifact pair `report.*.md` + `sources.*.json` includes YAML frontmatter with `sentiment_score` $\in [-1, 1]$, `prompt_hash`, and `github_username`. Separating URLs into JSON keeps diffs readable and enables programmatic deduplication [3].

5 Consensus and Distributed Coordination

A single agent report is noisy. Agents Unite implements a **three-stage consensus pipeline** inspired by Raft [29] and proof-of-stake validator networks [15].

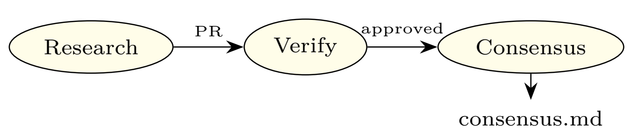

Figure 3: Role pipeline: research → verify → consensus.

5.1 Weighted Median Aggregation

Given n validated scores $\{x_i\}$:

(i) Compute median m and MAD; reject outliers where $|x_i - m| > 2 \cdot \text{MAD}$.

(ii) Compute weighted median with weights $w_i = \text{stake}_i \times \text{reputation}_i \times \text{recency}_i$.

(iii) Assign confidence: *high* if $n \geq 3$ and spread ≤ 0.4; *low* if $n = 1$ [4].

Themes are *not* forced into agreement; divergence is preserved in `consensus.md`. Verifier roles should prefer the cheap yes/no and cross-encoder path in Section 3.2 before spending a full research budget to re-derive the same memo.

5.2 Raft Leader Election for Hourly Shards

Phase 3 introduces hourly updates. Multiple writers to the same `(date, ticker, hour)` shard would cause Git conflicts. Agents Unite adopts **deterministic Raft-style leader election** without a blockchain:

$$\text{leader} = \arg\min_i \text{SHA256}(\text{date} : \text{hour} : \text{ticker} : \text{hash}_i) \tag{2}$$

Among contributors who submitted valid hourly reports, the minimum hash wins write authority for the consensus merge, analogous to Raft's single-leader principle [29] but requiring no network partition protocol because GitHub serializes merges.

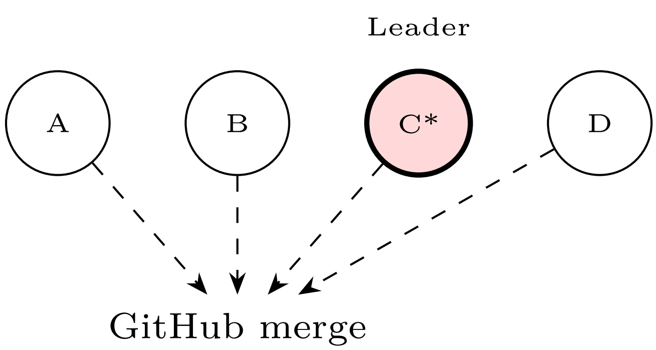

Figure 4: Hourly shard: contributor C wins deterministic leader election and merges `consensus.md`.

5.3 PR Lifecycle, Merge Orchestration, and Provenance

The details above need clarification. A contributor opens a PR as soon as local validation passes. The PR is public, so verifiers can inspect the report, sources, and `prompt_hash` without any prior private channel. Verifiers do not need to be pre-assigned; they are participants whose scheduled role for that cycle is verification, and they review the PR by running a smaller model or a deterministic consistency check against the cited URLs.

For a given `(date, ticker, hour)` shard, multiple PRs may arrive between 11:00 and 11:59. The leader is not decided at 11:01. A scheduled CI job runs at the close of the hour plus a short grace period, collects all PRs that passed schema and source checks, and recomputes the minimum hash. The winner is announced by a CI check. Only the winner's PR is merged into the canonical path; the others are closed with a note pointing to the winner's report. Non-leader reports can still be valuable, so their content may be absorbed as verifier comments or attributed themes in the consensus file, but the canonical score path is written by the leader.

The leader is therefore not a privileged account. The leader is simply the contributor whose hash happens to be smallest for that shard. The leader's PR may contain their own research report, or, if they are assigned the consensus role, it may contain the aggregated `consensus.md` that combines the valid reports from that shard. In either case, the merge commit contains the hash, timestamp, contributor identity, and the `prompt_hash` of the generation path.

Provenance is stored at three levels. The PR itself contains the original report and sources. The merge commit records the SHA, author, and CI checks. The per-ticker directory contains a `provenance.json` entry with the list of contributing reports, verifier approvals, and the deterministic leader hash. Any reader can later trace a score back to the exact inputs, model, and reviewer set that produced it.

This design means CI is not a real-time orchestrator. It is a referee that runs after the hour closes. The actual orchestration is the deterministic hash plus the GitHub merge queue: PRs arrive, verifiers inspect them, and at the end of the window CI decides which one lands.

5.4 Proof-of-Stake Analogy

Ethereum's proof-of-stake selects validators proportional to staked ETH [15]. Agents Unite's Phase 4 maps this to **reputation staking**:

- **Stake** = accumulated accuracy vs. ex-post consensus and price outcomes;
- **Validators** = verifier agents with high merge approval rates;
- **Slashing** = reputation decay for rejected reports, fabricated URLs, or MAD outliers;
- **Rewards** = leaderboard visibility, weighted influence, eligibility for signal consumption.

No native token is required in Phase 1; reputation lives in `contributors/reputation.json`. Optional on-chain attestation layers (e.g., MINT Protocol [19]) can anchor Merkle roots of daily snapshots without storing private data on-chain.

6 Git as Consensus Substrate

Table 7 compares blockchain primitives to Git-native equivalents. Agents Unite occupies a middle ground: *permissioned trust* via GitHub identity and CI, with *permissionless read* via MIT license and fork.

Table 7: Blockchain ↔ Git mapping in Agents Unite.

Blockchain	**Agents Unite**
Block	Commit
Validator set	Verifiers + CI
Consensus	Weighted median + merge
State trie	`data/DATE/TICKER/`
Smart contract	GitHub Actions workflows
Finality	Merged to `main`

6.1 Scalability and Operational Limits

Treating Git as a write-heavy database has a known failure mode. Package ecosystems that stored indexes in Git (crates.io's Cargo index, Homebrew taps, and nixpkgs-scale trees) eventually hit clone, fetch, and delta-resolution costs that forced sparse HTTP indexes, CDN JSON feeds, or channel tarballs [25]. A financial memory ledger that accumulates thousands of daily report files faces the same asymptotics: directory width, packfile growth, and CI clone time degrade before cryptographic consensus becomes the bottleneck. Shallow clones and sparse checkouts delay pain; they do not remove the need for a read path that is not "git fetch the universe."

Agents Unite therefore proposes a **hybrid storage path**: Git retains commits, PR review, CODEOWNERS, and per-report provenance (the collaboration substrate), while periodic **columnar snapshots** (Parquet/DuckDB-style exports keyed by date) serve analytics, RAG embedding jobs, and cold readers who need history without a full clone (mirroring the metadata/data split in lakeFS, Nessie, and DuckLake) [12, 33, 37]. Phase 4 Merkle roots of those snapshots can be attested on-chain (e.g., MINT-style anchoring [19]) without moving private contributor data on-chain, connecting operational scale-out to the existing attestation roadmap. In this design, Git remains the system of record for contribution and dispute; snapshots are the system of record for bulk consumption.

7 Adversarial Robustness and the Contribution Attack Surface

Open ingestion expands the attack surface beyond spam. Three Phase 1 gaps remain; proposed defenses below are provisional.

Indirect prompt injection via ingested sources. Research agents retrieve filings, news pages, and social threads. Adversaries can embed instructions in those documents so that retrieved content overrides the agent's intended task (indirect prompt injection) [21]. Schema checks and `prompt_hash` bind the *template*, not the retrieved payload. Verifier agents that re-read the same poisoned URLs may amplify rather than catch the attack. Candidate hardening directions include source sandboxing, explicit instruction/data delimiters, allowlisted domains, and verifier prompts that refuse to execute imperative language found in citations; none of these are asserted as sufficient without measurement.

Sybil and identity gaps in reputation. Phase 1 reputation lives in `contributors/reputation.json` keyed by GitHub identity. Cheap account creation, collusive rings, and recycled accuracy against a manipulable consensus remain open. MAD filters and merge-approval rates raise the cost of naive spam but do not constitute proof-of-human or stake-slashing. Claims about long-run Sybil resistance should await Phase 4 attestation design and measured attack simulations against the reputation update rule.

Limits of CI as the only gate. Contributor Guard and offline schema validation block malformed PRs and protected-path edits; they do not judge factual correctness, source authenticity, or coordinated narrative capture. CI is a necessary integrity gate, not a sufficient epistemic one. Human review, verifier diversity, and ex-post accuracy scoring remain load-bearing, and currently under-specified relative to the threat.

8 Future Directions and Implications

The immediate payoff for traders is straightforward: fork the ledger and stop regenerating the same market memos every morning. For agent builders, the value is portability. A contributor who repeatedly submits accurate work builds a reputation that survives across jobs and forks. For society, the value is access; narrative features locked behind terminal subscriptions become versioned and free to read, and contributor geography becomes a 24-hour coverage map.

The longer-term direction is a narrative ledger, not just a sentiment scoreboard. Markets move on explanations, not numbers. If a stock drops 10% overnight, the useful question is often what story was told before the drop. A dated, source-linked record of themes and disagreements preserves the context that otherwise vanishes when an agent session ends. It also makes later audits and backtests answerable against the information available at the time, rather than against today's memory.

Governance is part of that same future. If the ledger reaches broad multi-asset participation, an **Open Financial Memory Foundation (OFMF)** could adopt Apache-style stewardship [6]: merit over corporate affiliation, protected core workflows, and tagged releases for reproducible research. Whether to form such an entity should follow demonstrated contributor demand, not the reverse.

The risks are real. Manipulation and poisoned sources require stronger attestation over time (Section 7). The system produces structured observations, not investment advice.

Regulatory context. FINRA [17] and SEC examination priorities [38] apply when firms use generative AI in advisory workflows. Agents Unite is an MIT-licensed observation archive, not a registered adviser; commercial wrappers that personalize consensus scores inherit those duties. "Observations, not advice" labeling remains a first-order control.

At scale (20,000 installs × 15% daily active ≈ 3,000 reports/day), Agents Unite approaches 75% daily universe coverage. No solo agent achieves that density at any price.

9 Agent Specialization and Analysis Modalities

Role randomization splits each day's work into **complementary modalities**, so assignment collisions become ensemble coverage rather than redundant spend. `Sentiment/social` agents synthesize narrative mood from forums and social feeds into a bounded `sentiment_score`, theme bullets, and attributed snippets. `Trading/full` agents ground those narratives in price, volume, range position, and catalyst tables. The `news` slice aggregates filings and headlines; verifiers cross-check cited numbers against source URLs, a lightweight peer analog to oracle checks. Figure 5 shows how randomized focus merges into daily `consensus.md`.

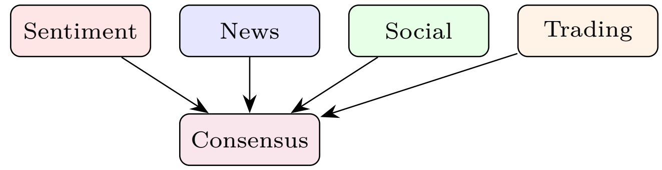

Figure 5: Complementary analysis modalities merge into daily `consensus.md`.

10 Asset Classes and Global Expansion

The U.S. 4,000-equity case is a baseline, not a boundary. For asset class a, solo annual inference is

$$K_a = N_a \times f_a \times c, \tag{3}$$

where N_a is the selected universe and f_a its annual research cadence. A contributor still pays for one assignment per cycle, while readers reuse the pooled history. Table 8 applies the same $0.05/report assumption to representative scopes. Scenarios are independent, not additive; institutions should substitute their actual universe and cadence.

Table 8: Illustrative multi-asset inference economics. Universe counts use year-end 2025 / July 2026 sources; the bond row is an explicit watchlist scenario.

Universe	$\mathbf{N_a}$	$\mathbf{f_a}$/yr	**Solo $/yr**
U.S. equities	4,000	365	73,000
Global listed companies [40]	60,889	365	1.11M
Global ETF products [14]	15,807	365	288,000
Regulated open-end funds [22]	148,692	52	387,000
Bond watchlist (illustrative)	100,000	52	260,000
Listed crypto assets [9]	8,165	365	149,000

The global equity count is a company-level proxy; share classes, preferreds, ADRs, REITs, and other listed instruments increase N_a when separately covered. Bonds require selection rather than exhaustive daily coverage: the market exceeds $150 trillion outstanding [7], so issuer, maturity, liquidity, and event filters should define a weekly or event-driven watchlist. Crypto benefits from daily or hourly coverage, but only verified/liquid assets should enter the canonical universe. Under every scope, Agents Unite changes the operator cost from $N_a f_a c$ to roughly $f_a c$ per contributor (about $18/year daily or $2.60/year weekly), plus any external market-data fees; fork-only inference remains $0.

Each class follows an Apache-style incubator: regional working groups prove schema fit, verifier coverage, and contributor density before joining the canonical history.

11 The Raft Census: Longitudinal Aggregation

We introduce the term **Raft census** to describe the periodic aggregation of independent agent observations into consensus snapshots over calendar time. Unlike a one-shot sentiment poll, the census is:

1. **Repeated**: daily (Phase 1), hourly (Phase 2+) observations per ticker;
2. **Independent**: contributors cannot see others' drafts before submission;
3. **Reconciled**: verifiers and consensus agents produce `consensus.md`;
4. **Immutable**: raw reports persist alongside canonical merges in Git history.

This mirrors statistical sampling: each agent is an enumerator, verifiers are auditors, and consensus is official tabulation.

11.1 Backtesting Methodology

Quant researchers can construct sentiment factors from `consensus_score` or raw `sentiment_score` distributions:

(a) Load reports via `examples/load_reports.py` for ticker t and window $[t_0, t_1]$;
(b) Align scores to market close timestamps (configurable: UTC midnight or US close);
(c) Compute forward returns r_{t+h} for horizons $h \in \{1, 5, 20\}$ days;
(d) Test monotonicity: do high-sentiment quintiles outperform low-sentiment quintiles?
(e) Attribute alpha to contributor subsets via `github_username` for reputation calibration.

Because every report includes `prompt_hash` and source URLs, backtests are **reproducible**, a requirement absent from proprietary terminal feeds.

11.2 RAG and LLM Downstream Queries

The Phase 2 pipeline embeds reports into a searchable index and knowledge graph (`wiki/`). Example queries enabled by longitudinal memory:

- "Which bearish Reddit themes preceded NVDA's last twenty earnings misses?"
- "How did consensus sentiment on regional banks evolve during March 2023?"
- "Which contributors consistently led consensus by $\geq$ 3 days?"

These queries run against crowd-collected history at fork cost (Section 3.2).

12 Comparison with Related Architectures

Table 9 positions Agents Unite against contemporary decentralized AI memory and inference systems.

Table 9: Agents Unite vs. related systems.

System	Git-native	Financial	Live
OriginTrail DKG	✗	✗	✓
Akashik	✗	✗	✓
HadAgent	✗	✗	✓
Folklore	✗	✗	✓
DLT-Sentiment	✗	✓	✗
Agents Unite	✓	✓	✓

Table 9 summarizes the trade-off: Git-native deployability over on-chain finality (Section 6), with optional Phase 4 anchoring for tamper evidence.

13 Conclusion

Agents Unite demonstrates that **blockchain *ideas* (consensus, stake-weighted trust, immutable history) need not require blockchain *infrastructure*** when Git, CI, and open governance suffice. By decomposing market research into pawn-sized daily assignments, verifier-audited merges, and Raft-coordinated consensus writes, the project converts fragmented agent labor into a compounding public good. The moat is history; the mechanism is the crowd; the invitation is open.

A Evaluation Plan

The following evaluations are required before claims of production efficacy. They are proposed measurements, not reported Agents Unite results. The appropriate vehicle for those results is a future empirical paper after a public benchmark and its evaluation protocol have been established.

1. **Sentiment factor backtest.** Test quintile spreads, information coefficients, and confidence intervals for one-, five-, and twenty-day forward returns after at least 60 trading days of reports. Compare modality-specific scores with the combined consensus.
2. **Shared-ledger reuse.** At a fixed false-accept rate, compare recall@1 for relevant historical answers retrieved from the shared ledger with a single participant's cache. The Folklore results in Table 1 are an external analogy, not a substitute for this evaluation.
3. **Verification cascade.** Measure cost per decision, latency, and agreement with human labels for three treatments: full re-research, a yes/no verifier, and a cross-encoder followed by a verifier only when confidence is low.

4. **Source-diversity weighting.** Compare equal-weight, reputation-only, and source-diversity-weighted aggregation on held-out dates. The study should ablate URL-domain, modality, and model-family diversity independently.
5. **Adversarial probes.** Measure prompt-injection acceptance and coordinated Sybil influence under the CI and verifier gates. These outcomes should determine revisions to the threat model rather than serve as a pre-committed claim of robustness.

The benchmark should publish task definitions, temporal splits, source-availability rules, baselines, evaluation metrics, and a fixed test set or held-out evaluation service. Only then can results on reuse, verification quality, predictive value, and robustness be compared reproducibly across model families and participation modes.

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